

WEARABLE ACCELEROMETER FOUNDATION MODELS FOR HEALTH via KNOWLEDGE DISTILLATION

This Paper builds heavily off of the previous paper from 2024 called "Large Scale foundation models for wearable bio signals". In that paper they used contrastive learning to learn good PPG encodings via SSL that was then very helpful for downstream tasks.

But PPG recording is a "high fidelity" recording so not all wearables can record it as it's very energy intensive.

But "low fidelity" sensors like accel can in periods of no movement capture minute movements of the body like heart function this is known as "ballistocardiogram"

SO, the goal is to use these
"no motion" accel segments to predict
downstream health things.

They approach this in 3-ways in the
paper:

- 1) Accel-Supervised
(inputs accel, outputs are supervised with
downstream health task)
- 2) SSL on Accel (contrastive or MAE)
then
Accel-Supervised
masked autoEncoder
- 3) SSL on PPG
then
representational
knowledge
distillation to
Accel model
then
Accel-Supervised
- 4) SSL on PPG and Accel
(at the same time with contrastive loss)
then
Accel-Supervised

They do this and find that #3 works the best.

Now for the details:

- They train the teacher model with MAE where the 4-channel input gets projected via a learnable tokenizer then we drop 80% of tokens then passed through model and loss is to reconstruct 4-channel missing pixel values in the PPG.

↳ NOTE: Authors make claim but don't verify. That MAE is better for less-noisy biosignals like PPG + CL(contrastive) is better for noiser biosignals.

- PPG-MAE outperforms PPG-CL
- Accel-CL outperforms Accel-MAE
- ↳ This pattern is consistent throughout paper

- Then they freeze the teacher + distill to the student accel-model. (similar to the CLIP model Idea with image text pairs)

- Steps to distill:

- take input accel & PPG Pair
and add random noise to them

↳ note the noise addition is important
they do ablation study and find
that embeddings are worse quality
if you don't add noise

- Pass Both through their respective
encoders & use separate Learnable
MLP heads to project down to
128-D

↳ they find without this step you
get representational collapse

- They use INFONCE LOSS as a
weighted loss from teacher $\rightarrow$ student
& student $\rightarrow$ teacher

↳ they try different weights.
& find best results from
 $\text{weight}(t \rightarrow s) = 1$
& $\text{weight}(s \rightarrow t) = 0$

Results & details:

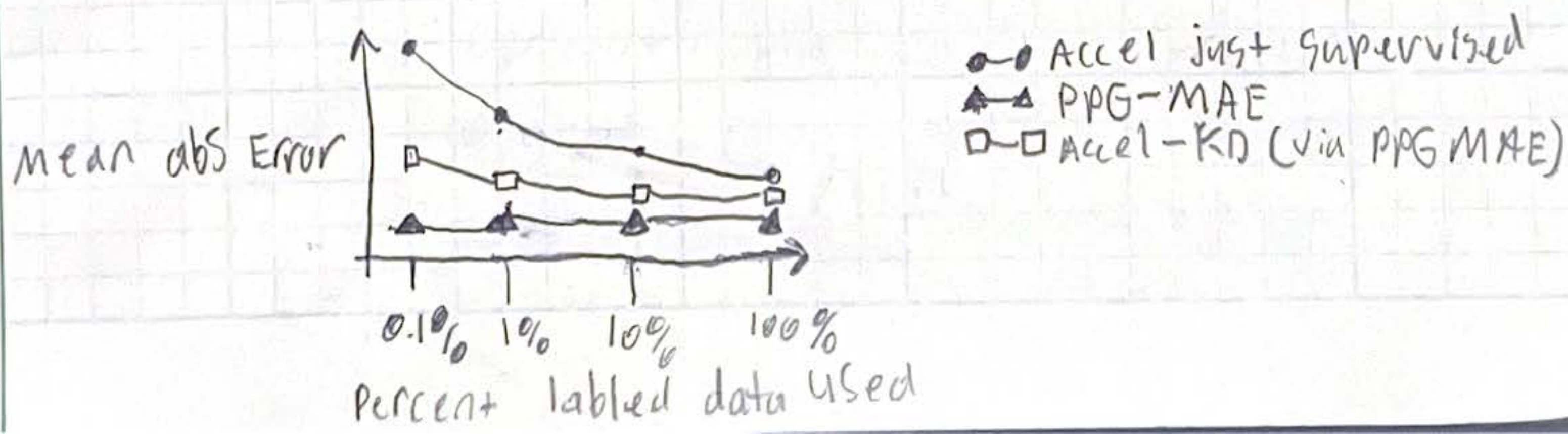
- We presented their results to start now we will expand.

1) they test if Accel-KD (via PPG-MAE) & PPG-MAE embeddings are close using t-sne & they are close. Then they test if Accel-MAE embeddings are close to PPG-MAE embeddings (so No-Knowledge distillation) and they are not

2) Knowledge Distillation to Accel model allows it to perform much better as a pose to just pre-training on Accel

3) Pre-training allows for robust results across all amounts of labeled data 0.1% - 100%

↳ but Pretrain PPG then Knowledge distillation results in best Accel Model



4) Cross-model distillation also allows for model compression

+ When distilling from PPG-MAE to Student Accel model they can make student 5x smaller than default and still get better results than baseline (Accel-MAE + Accel-CL)

5) if the models of the student are Efficient-net or transformers, regardless the results are very similar

6) Why not train the teacher and student at the same time? (Like option #4)

- you get worse results.

- Their Hypothesis why

↳ PPG is info rich + Accel is noisy

↳ So, contrastive loss allows PPG encoder to be lazy and learn worse Embeddings

↳ Since They have different "information content" the simultaneous training degrades representations, compared to learning from good teacher.

Note: IF you look at figure 2. Then you might be confused because how could model that got pre-training & labeled data (Accel-MAE) do worse than model with just labeled data? That is because they don't do any fine-tuning the Backbone weights are always frozen during down stream tasks & they just do linear probing